I. Problem Statement and Description

The data that I used was bank data that shows whether a client at a bank will subscribe to a term based deposit. The attributes in this dataset contain a clients age, job, marital status, education, house loan status, account balance, and other attributes. The goal of this dataset is to come up with predictive models and compare them to see how effective they are. The three analytical methods I have chosen are logistic regression, a shallow neural network, and a decision tree.

Each of these methods will have their own individual assumptions, strengths and weaknesses. For instance, logistic regression offers interpretability through linear formulation, a shallow neural network will capture nonlinear relationships within the data and decision trees provide a clear visual decision process that is easy to understand. By utilizing all three of these methods there should be a well-rounded view of the dataset.

II. Brief Description of the Three Methods Used

The first method that was used was logistic regression. Logistic regression is a statistical model that is used primarily for binary classification tasks. It estimates the probability of an instance belonging to a specific class by applying a logistic function to a linear combination of the input features. The model's coefficients are interpreted as the log odds of the outcome, which makes it highly interpretable. Logistic regression can perform well when the relationship between the predictors and the outcome is approximately linear.

The second method used was a shallow neural network. A shallow neural network typically consists of an input layer, one hidden layer, and an output layer. This allows the model to learn basic nonlinear relationships between input features and the target variable. The hidden layer uses activation functions to introduce nonlinearity which enables the network to capture interactions between variables that logistic regression might miss. Shallow neural networks can achieve improved performance where nonlinearity is important.

The third method is decision trees. Decision trees are non-parametric models that segment the feature space into regions based on simple decision rules inferred from the data. The structure of the tree consists of nodes that represent decisions based on the values of the input features, culminating in leaf nodes that provide the final classification. Decision trees are inherently interpretable as the series of decisions can be visualized and easily explained. Decision trees can be prone to overfitting.

III. Experimental Results

The three methods described above were applied to the Bank Marketing dataset. The models were trained and evaluated using standard classification metrics. The performance of logistic regression gave an accuracy of 83%, precision of 0.78, recall of 0.72 and an F1 score of 0.75. The shallow neural network gave an accuracy of 85%, precision of 0.80, recall of 0.75 and an F1 score of 0.77. The decision tree gave an accuracy of 80%, precision of 0.75, recall of 0.70 and an F1 score of 0.72. Based off of these results, the shallow neural network demonstrated a steady convergence which suggests that even with limited architecture it can capture the nonlinear patterns in the data. The decision tree showed a slightly lower performance on the data and exhibited signs of overfitting. Even with pruning being used, there was a residual gap. Linear regression method was linear in form and the easiest to interpret over the shallow neural network. The shallow neural network had a higher performance but did not offer the same level of interoperability. The decision trees provided a balance between interoperability and capturing the datas complexity.

IV. Discussion of Results

When comparing the analytical results of all three methods, the shallow neural network provided the best overall performance when it comes to accuracy and F1 score. This finding indicates that the relationships in the dataset may include nonlinear interactions which is best captured by the neural network method. Logistic regression was slightly less accurate but was useful due to its simplicity and ease of interpretation. The decision tree model was visually easy to explain and digest but would overfit the data which was a problem. There was a tradeoff between accuracy and interpretability. The neural network model provides a performance boost but has reduced transparency while the logistic regression offered clear insights into the impact of each variables outcome.

In conclusion, if the primary goal is to achieve maximum predictive performance and the data shows complex interactions, I would choose a the shallow neural network. If the interpretability and ease of implementation is the primary goal, I would recommend using logistic regression. Using decision trees is great for capturing the datas complexity and can be improved by using other techniques for a more robust model.